

# MERN STACK INTERNSHIP – DAY 15 ASSIGNMENT REPORT

## Company

Insta Dot Analytics

## Submitted By

Name: Chayan Soni

Submission Date: 17-06-2026

## FOR SOURCE CODE:

**GIT HUB** :- <https://github.com/chayan-soni/MERN-Internship>

## Internship Day 15 – Order Management Module Development

### Objective

The objective of Day 15 was to develop and integrate a complete Order Management Module in the MERN E-Commerce application. This included designing MongoDB schemas for Orders and Order Items, creating REST APIs for order operations, integrating the checkout process with the backend, displaying order history for users, and testing all APIs using Postman.

## Tasks Performed

### 1. Created Order and Order Item Schemas

Designed MongoDB schemas using Mongoose to store order-related information.

#### *Order Schema Fields*

- User ID
- Order Items
- Shipping Address
- Payment Method
- Total Amount
- Order Status
- Order Date

- Created At / Updated At

#### *Order Item Schema Fields*

- Product ID
- Product Name
- Product Price
- Quantity
- Product Image

The schemas were structured to maintain data consistency and establish relationships between users, products, and orders.

## 2. Developed Order Management REST APIs

Implemented RESTful APIs to handle order operations within the application.

#### *Place Order API*

**Purpose:** Create a new order after successful checkout.

**Features:**

- Accepts cart details from frontend.
- Validates user and cart information.
- Stores order data in MongoDB.
- Generates order record successfully.

#### *View Orders API*

**Purpose:** Retrieve order history of a logged-in user.

**Features:**

- Fetches all orders associated with the user.
- Displays order status and order details.
- Returns data in structured JSON format.

#### *Cancel Order API*

**Purpose:** Allow users to cancel eligible orders.

**Features:**

- Verifies order ownership.

- Updates order status to "Cancelled".
- Prevents unauthorized cancellation requests.

### 3. Checkout Page Integration

Connected the React Checkout page with backend Order APIs.

#### *Implementation Details*

- Collected cart items and shipping information.
- Sent order details to backend
- Handled API responses.
- Displayed order placement confirmation messages.
- Cleared cart after successful order creation.

This integration enabled users to complete purchases directly from the frontend application.

### 4. Implemented Order History Page

Developed a dedicated Order History section for authenticated users.

#### *Features*

- Displays all previous orders.
- Shows order ID and order date.
- Displays total amount and order status.
- Provides easy tracking of user purchases.
- Fetches real-time data from backend APIs.

### 5. API Testing Using Postman

Tested all Order Management APIs using Postman.

#### *Tested APIs*

1. Place Order API
2. View Orders API
3. Cancel Order API

### *Validation Performed*

- Successful API execution.
- Request and response verification.
- Error handling validation.
- Authentication checks.
- Database record verification.

All APIs produced the expected responses and updated the database correctly.

### Technologies Used

- MongoDB
- Express.js
- React.js
- Node.js
- Mongoose
- Postman
- Git & GitHub

### Challenges Faced

- Designing relationships between Orders and Products.
- Managing order status updates efficiently.
- Handling checkout data transfer between frontend and backend.
- Ensuring authenticated access to order-related APIs.

### Learning Outcomes

Through this task, I learned:

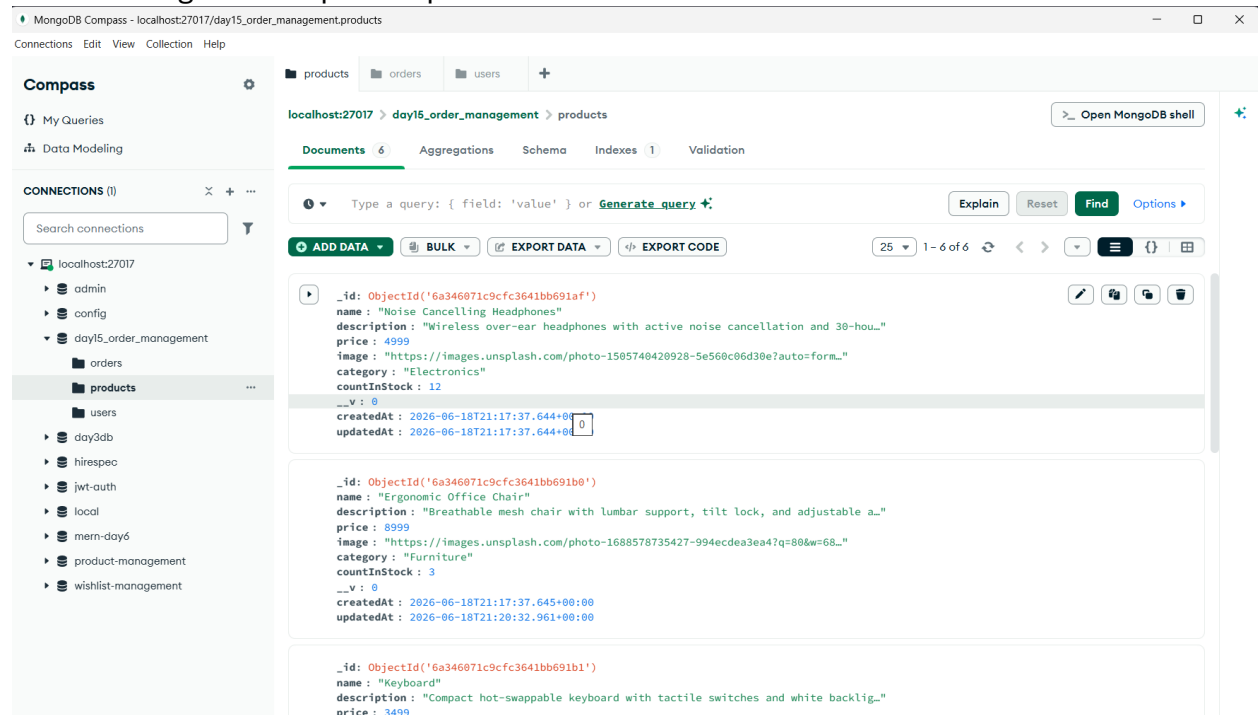
- MongoDB schema design for order management systems.
- Building scalable REST APIs using Express.js.
- Managing order workflows in an e-commerce application.
- Integrating frontend checkout functionality with backend services.
- Testing and validating APIs using Postman.
- Handling user-specific data securely.

## GitHub Submission

The updated source code containing the Order Management Module was pushed successfully to the GitHub repository.

## Screenshots Section

Data in MongoDB compass of products:



## Data in MongoDB compass of users:

MongoDB Compass - localhost:27017/day15\_order\_management.users

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

- admin
- config
- day15\_order\_management
  - orders
  - products
  - users**
- day3db
- hirespec
- jwt-auth
- local
- mern-day6
- product-management
- wishlist-management

localhost:27017 > day15\_order\_management > users

Documents 1 Aggregations Schema Indexes Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

ADD DATA BULK EXPORT DATA EXPORT CODE

25 1-1 of 1

```
{
  "_id": ObjectId('6a345d7289287dc84f832fe1'),
  "name": "Chayan Soni",
  "email": "chayanson104@gmail.com",
  "password": "$2a$10$0R2BSNfBLGRMtGTxrR8CEus53yPkjq1xJGTthAygDtTrhmUKjzW5",
  "createdAt": 2026-06-18T21:04:50.816+00:00,
  "updatedAt": 2026-06-18T21:04:50.816+00:00,
  "__v": 0
}
```

## Orders:

MongoDB Compass - localhost:27017/day15\_order\_management.orders

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

- admin
- config
- day15\_order\_management
  - orders**
  - products
  - users
- day3db
- hirespec
- jwt-auth
- local
- mern-day6
- product-management
- wishlist-management

localhost:27017 > day15\_order\_management > orders

Documents 1 Aggregations Schema Indexes Validation

Type a query: { field: 'value' } or [Generate query](#)

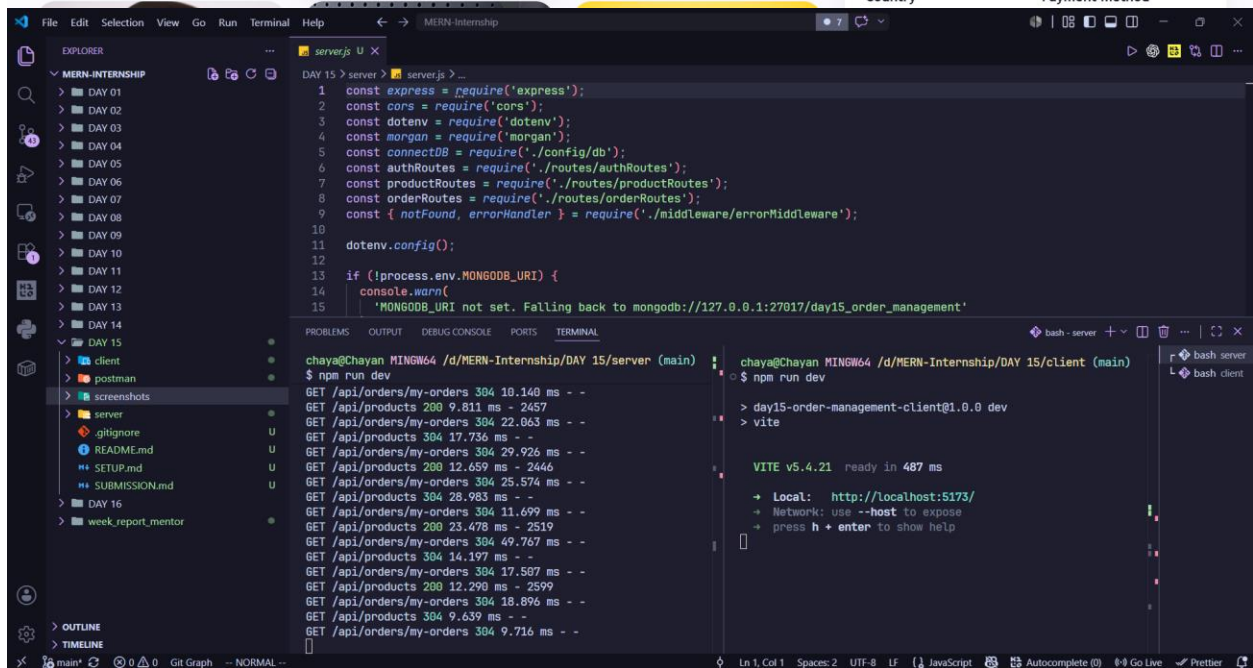
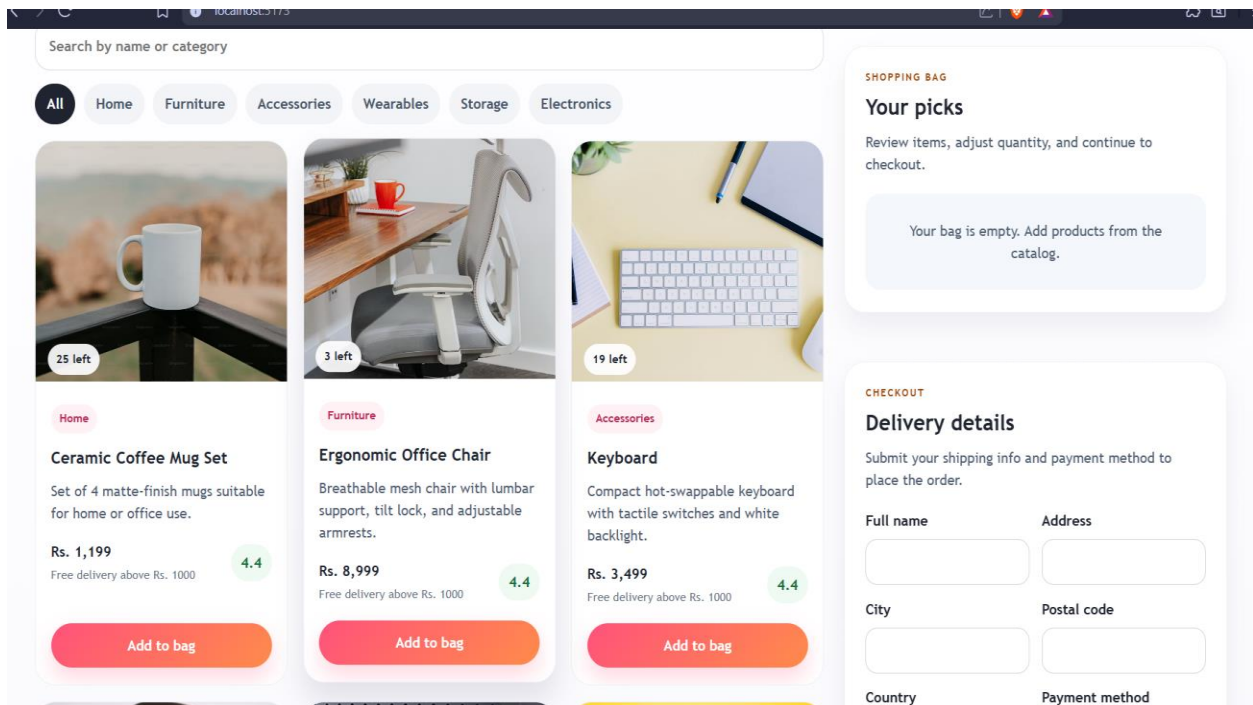
Explain Reset Find Options

ADD DATA BULK EXPORT DATA EXPORT CODE

25 1-1 of 1

```
{
  "_id": ObjectId('6a346128c9cfc3641bb691c0'),
  "user": ObjectId('6a345d7289287dc84f832fe1'),
  "orderItems": Array (2),
  "shippingAddress": Object {
    "paymentMethod": "Cash on Delivery"
  },
  "itemsPrice": 48494,
  "taxPrice": 8728.92,
  "shippingPrice": 0,
  "totalPrice": 57222.92,
  "status": "Placed",
  "createdAt": 2026-06-18T21:20:32.952+00:00,
  "updatedAt": 2026-06-18T21:20:32.952+00:00,
  "__v": 0
}
```

## api\_responses:



ORDERS

## Recent orders

Track previously placed orders and cancel any active order.

Placed

19/06/2026, 03:16:34Rs. 14,747.64

### Order #B691E5

Ergonomic Office Chair

1 x Rs. 8,999

Keyboard

1 x Rs. 3,499

Ship to: Chayan Soni, 1140-G scheme number 114, vijay nagar, indore

Payment: Cash on Delivery

Cancel order

Placed

19/06/2026, 02:50:32Rs. 57,222.92

### Order #B691C0

Ergonomic Office Chair

5 x Rs. 8,999

Mechanical Keyboard

1 x Rs. 3,499

Ship to: chayan , 1140 vijay nagar , indore

Payment: Cash on Delivery

Cancel order



CHECKOUT

Office Chair

Remove

Keyboard

Remove

## Delivery details

Submit your shipping info and payment method to place the order.

Full name

Address

Chayan Soni

1140-G scheme numb

City

Postal code

indore

4542010

Country

Payment method

India

Cash on Delivery

Place order

SHOPPING BAG

## Your picks

Review items, adjust quantity, and continue to checkout.



Ergonomic Office Chair

Rs. 8,999

Qty

1

Remove



Keyboard

Rs. 3,499

Qty

1

Remove

|          |               |
|----------|---------------|
| Items    | Rs. 12,498    |
| Shipping | Rs. 0         |
| Tax      | Rs. 2,249.64  |
| Total    | Rs. 14,747.64 |